

Cs610

Computer Network

Most Repeated Subjective Midterm

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|--------------------------|----------------------|-----------------------|
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Cs601 Midterm Repeated Question

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Short Question

Question No 1

Define Vector-Distance Algorithm?

Answer

Packet switches wait for next update message and they iterate through entries in message. If entry has shortest path to destination, insert source as next hop to destination and record distance as distance from next hop to destination plus distance from this switch to next hop.

Question No 2

CLASSIFICATION OF NETWORKS

Answer

Computer networks are classified by four factors which are as follow:

- 1) BY SIZE:
- 2) BY CONNECTIVITY:
- 3) BY MEDIUM:
- 4) BY MOBILITY:

Question No 3

Define static and dynamic routing?

Answer

Static Routing

Static routing is the simplest form of routing, but it is a manual process.

Dynamic Routing

Dynamic routing protocols are supported by software applications running on the routing device (the router) which dynamically learn network destinations and how to get to them and also advertise those destinations to other routers.

Question No 4

How many uses of CRC hardware Component?

Answer

CRC uses just two hardware components:

- Shift register
- Exclusive OR (XOR unit)

Question No 5

What is 10base T?

Answer

One of several adaptations of the Ethernet (IEEE 802.3) standard for Local Area Networks (LANs). The 10Base-T standard (also called Twisted Pair Ethernet) uses a twisted-pair cable with maximum lengths of 100 Meters.

Question No 6

What is Multicasting?

Answer

It works like broadcasting however it does not forward frames automatically to the CPU. The interface hardware is programmed in advance to accept certain frames that have multicast address as the destination address.

Question No 7

Difference between EXPLICIT FRAME TYPE and IMPLICIT FRAME TYPE?

Answer

IMPLICIT FRAME TYPE:

In implicit frame the receiver must infer from frame data.

EXPLICIT FRAME TYPE:

In this type the identifying value is included with frame describes types of included data.

Question No 8

What is FDDI?

Answer

Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are:

- It uses fiber optics between stations and transmits data at 100Mbps.
- It uses pair of fibers to form two concentric rings.

Question No 9

Defines simplex and full duplex connection?

Answer

Some connection-oriented technologies provide full duplex while other allow on simplex connection. To communicate using a simplex design a pair of computers must establish two connections one from computer A to computer B and another from computer B to A.

Question No 10

How WAN capable to handle many computers?

Answer

A Wide Area Network (WAN) is a telecommunication network that covers a broad area.

Question No 11

Briefly explains the working structure of thick Ethernet. Marks 2

Answer

It uses thick coax cable. AUI cable (or transceiver or drop cable) connects from NIC to transceiver. AUI cable carries digital signal from NIC to transceiver. The transceiver generates analog signal on coax cable. The wires in AUI carry digital signals power and other control signals. Thick Ethernet also requires terminators to avoid signal reflectance.

Question No 12

What type of topology used in Ethernet?

Answer

Traditional Ethernet employs a bus topology, meaning that all devices or hosts on the network use the same shared communication line.

Question No 13

Define Auto Negotiation?

Answer

100Base-T technology is backward compatible and allows the participants to negotiate a speed when connection is established. This process is known as auto negotiation.

Question No 14

Difference between LAN and WAN?

Answer

In LAN network occupies the smaller area like a room a floor or a building.

In WAN, network occupies larger areas like cities & countries. Internet is a Wide Area Network.

Question No 15

Who examine performance of network?

Answer

Performance is probably the most closely monitored operational feature by users in the network environment

Question No 16

How can we compute shortest path in WAN?

Answer

We use Dijkstra's algorithm to compute shortest path from each node to every other node.

Question No 17

Where are destination and source address located in frame?

Answer

The packet header includes destination and source addresses.

Question No 18

If 100mbps Ethernet NIC is connected to a hub which supports 10mbps, what will be the data rate in this scenario?

Answer

This has two consequences. First, the hub does not necessarily create a broadcast medium. Second, there are no collisions between traffic on 10 Mbps ports and 100 Mbps ports.

Question No 19

In connection oriented network, which type of connection identifier is used? Support your answer with example?

Answer

ATM uses the concept of connection-oriented networking.

Question No 20

Network administrator needs audio and video service for his network. He has to send or receive data at same data rate. Which type of ATM quality service specification would be applied there? Give example?

Answer

To facilitate the transport of audio and video data across emerging Asynchronous Transfer Mode (ATM) networks, a simple, low cost, audio/video ATM appliance, the AVATAR, has been developed. This appliance is capable of handling uncompressed bidirectional audio and NTSC video connections.

Question No 21

Network administrator wants to build a tunnel between sites offices. How will he build using private virtual network?

Answer

To establishes a private network that can send data securely between these two locations or networks through a "tunnel." A VPN tunnel connects the two PCs or networks and allows data to be transmitted over the Internet as if it were still within those networks.

Question No 22

Define bridge?

Answer

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. A typical bridge has two NICs, a CPU a memory and a ROM. It only runs the code stored in its ROM.

Question No23

Disadvantages of Repeaters?

Answer

Repeaters do not recognize frame formats, they just amplify and retransmit the electrical signal. If a collision or error occurs in one segment, repeaters amplify and retransmit also the error onto the other segments.

Question No 24

What is meant by Bridges STARTUP and STEADY State?

Answer

When a bridge first boots the address lists are empty (start up state). The bridge forwards frames to the other segment if it can not find its destination address in its lists.

After some time when the bridge has received at least one frame from every computer, it has the lists built (steady state) it forwards frames as far it is necessary.

Question No 25

How can Switched Virtual Network be established?

Answer

ESTABLISHING AN SVC:

The computer sends a connection request to the switch to which it is attached. Software in the switch finds a network path to the destination and sends along the connection request.

Each pair of switches in the path communicates to choose a VPI/VCI for their tables. Once the connection is established by the destination, a message is sent back to the originating computer to indicate the SVC is ready.

If any switch or the destination computer does not agree to setting up the VC, an error message is sent back and the SVC is not established

Question No 26

What is the concept of store and forward technology?

Answer

STORE AND FORWARD

Data delivery from one computer to another is accomplished through store and forward technology. In this technology packet switch stores incoming packet and also forwards that packet to another switch or computer. For this purpose packet switch has internal memory into which it can hold packet if outgoing connection is busy. Packets for each connection held on queue

Question No 27

Give a comparison of wiring Schemes?

Answer

The wiring schemes are compared as follows:

Separate transceiver allows computers to be powered off or disconnected from network without disrupting other communication.

Transceiver may be located in an inconvenient place, so finely malfunction transceiver can be hard.

In other case, thin coax cable takes minimum of cable. Disconnecting one computer (on one loose connection) can disrupt entire network.

Hub wiring centralizes electronics and connections. It makes management easier. Bottom line 10Base-T is most popular because of lowest cost.

Long Questions

Question No 1

Define 802.11 Wireless lans and CSMA?

Answer

IEEE 802.11 is standard wireless LAN that uses radio signals at 2.4GHz. Its data rate is 11Mbps. The older devices use radio signals at 900MHz and data rate of 2Mbps. Bluetooth specifies a wireless LAN for short distances. It uses shared medium and radio waves instead of coaxial cable.

Question No 2

Describe permanent virtual circuits (PVC)?

Answer

ATM can provide customers with virtual circuits that look like traditional leased digital circuits. Such permanent virtual circuits (PVC) last as long as the customer pay the periodic fee for its use. The forwarding tables are automatically restored after power of equipment failure. The forwarding table entries for such permanent VC's are statically configured, the terms used by Telco's for this is provisioning.

Provisioning requires two steps:

1. To determine a complete path (that is, identify the switches that will be used).
2. To choose appropriate VPI/VCI for each step in the path, and configure each adjacent pair of switches (easy, since each switch rewrites the VCI/VPI).

Question No 3

How can a bridge know whether to forward frames?

Answer

The bridges configure themselves automatically to decide which bridge will forward broadcast frames and which bridge will not.

The bridges communicate with each other on the network and use Distributed Spanning Tree (DST) algorithm to decide which bridge will not forward frames if a cycle occurs.

Question No 4

What is the concept of packet switches?

Answer

To span long distances or many computers, networks must replace shared medium with packet switches. Each switch moves an entire packet from one connection to another. That's why they are called packet switches. A packet switch consists of a small computer with network interfaces, a memory and a program dedicated to packet switching function.

Question No 5

Write a note on Weight Graph?

Answer

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight (sum of the weight with all edges). It should be noted that the shortest path is not necessarily with fewest edges (or hops)

Question No 6

Different b/w Cells and Packets?

Answer

ATM designers chose cells over packets because of the following reasons:

- Cells are not variable length and memory management for them is simpler. Handling variable length packets leads to memory fragmentation.

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- Variable length packets require hardware to accommodate the largest possible packet, and thus to detect the end of the packet. With cells bits can just be counted as they arrive.
- The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission. QoS can't be guaranteed with variable length packets as easily as it can with fixed length cells.

Question No 7

Why ATM design chose cells over packet?

Answer

- Cells are not variable length and memory management for them is simpler. Handling variable length packets leads to memory fragmentation.
- Variable length packets require hardware to accommodate the largest possible packet, and thus to detect the end of the packet. With cells bits can just be counted as they arrive.
- The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission. QoS can't be guaranteed with variable length packets as easily as it can with fixed length cells.

Question No 8

What type of switching technique is used in WAN?

Answer

WANs can use circuit-switching or packet-switching techniques. To span long distances or many computers, networks must replace shared medium with packet switches. Each switch moves an entire packet from one connection to another. That's why they are called packet switches.

Question No 9

Star Textile has huge network of systems and switches. A task is given to IT Officer that if some link goes the entire network works smoothly. Which routing method will be use? 5 marks

Answer

Distributed routing method is better for this purpose because Distributed routing relies on each node to compute its own routing table and build the required connections with its neighbors.

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Ideally, the network operation, status, and architecture of each node is transparent. Distributed routing is more flexible than centralized routing because each node handles its own routing. The result is often improved system performance.

Question No 10

We have two satellite locations, at first location one bridge is configured and at second location 2 bridges are configured. Which location performed will be faster? Give reason. (3 marks)

Answer

Satellite locations with two bridges configured will performed faster. The brigade also had to provide power to virtually all of its East Timor assets. Setting up the satellite system, for example, required supplying power to both the communications station and the users' computers. A bridge that has multiple ports is known as a networking switch. Both bridges and switches are capable of directing traffic to specific network addresses instead of broadcasting the data to all devices on a network segment. This functionality makes the bridge or switches a more advanced networking device over a hub or repeater.

Question No 11

Define Data Stuffing?

Answer

In general to distinguish between data being sent and control information such as frame delimiters network systems arrange for the sending side to change the data slightly before it is sent because systems usually insert data or bytes to change data for transmission, the technique is known as Data Stuffing.

Question No 12

How many Categories of Wires?

Answer

CATEGORIES OF WIRES:

Cable used for wiring should match the following:

- The intended data rate

- The distance between devices
- The amount of em-noise
- Anticipated future needs
- Cost

Question No 13

What is Simplex and full Duplex Connection?

Answer

SIMPLEX & FULL DUPLEX CONNECTION:

Some connection-oriented technologies provide full duplex while other allow on simplex connection. To communicate using a simplex design a pair of computers must establish two connections one from computer A to computer B and another from computer B to A.

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آپ سب سے گزارش ہے پلیز دعاؤں میں یاد رکھیں
مزید ایسی پوسٹ حاصل کرنے کیلئے ہمارا یوٹیوب چینل سبسکرائب کریں
اللہ پاک آپ سب کو اور مجھے اپنے مشن میں کامیابی عطا فرمائے۔۔۔ آمین

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